RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-01

Single Correct Question

1. The value of the determinant $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix}$ is equal to -

(A) $a^3 - b^3$ (B) $a^3 + b^3$ (C) 0 (D) none of these

MDT213

2. Which of the following is correct?

(A) $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = 17$ (B) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 12$ (C) $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 46$ (D) $\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 4$

MDT214

3. If the minor of three-one element (i.e. M_{31}) in the determinant $\begin{vmatrix} 0 & 1 & \sec\alpha \\ \tan\alpha & -\sec\alpha & \tan\alpha \\ 1 & 0 & 1 \end{vmatrix}$ is 1 then find the value of $\alpha.(0 \le \alpha \le \pi)$

(A) 0 (B) $\frac{3\pi}{4}$ (C) π (D) All of these

Rankers Academy JEE

MDT215

4. If $f(x) = \begin{vmatrix} x-1 & x+1 \\ x-2 & x+2 \end{vmatrix}$, then find the value of $\begin{vmatrix} f(-1) & f(-2) \\ f(1) & f(2) \end{vmatrix}$.

(A) 0 (B) 1 (C) 3 (D) 4

MDT216

Comprehension Type Question

Comprehension (For Q.5 to Q.7)

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ \ell & m & n \end{vmatrix}$

M_{ij} denotes the minor of an element in i^{th} row and j^{th} column

C_{ij} denotes the cofactor of an element in i^{th} row and j^{th} column

5. The value of $p.C_{21} + q.C_{22} + r.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

MDT217

6. The value of $x.C_{21} + y.C_{22} + z.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

MDT218

For More Join: @IITJEE_Advanced

7. The value of $q.M_{12} - y.M_{22} + m.M_{32}$ is

(A) 0 (B) $-\Delta$

(C) $x.M_{21} - y.M_{22} + z.M_{23}$ (D) Both (B) & (C)

MDT219

Subjective Type Question

8. Calculate the value of the determinant $\begin{vmatrix} 5 & -3 & 7 \\ -2 & 4 & -8 \\ 9 & 3 & -10 \end{vmatrix}$

MDT220

9. Find the value of 'k', if $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & k & 2 \end{vmatrix} = 4$

MDT221

10. Find minors & cofactors of elements '6', '5', '0' & '4' of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 6 & 5 & 7 \\ 3 & 0 & 4 \end{vmatrix}$.

MDT222

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	B	C	D	A	A	C	D	–98	0	

10. (minors : 4, –1, –4, 4 ; cofactors : –4, –1, 4, 4)

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-02

Single Correct Question

1. If $D=\begin{vmatrix}\alpha & \beta\\ \gamma & \delta\end{vmatrix}$, then $\begin{vmatrix}2\alpha & 2\beta\\ 2\gamma & 2\delta\end{vmatrix}$ is equal to -

(A) D (B) $2D$ (C) $4D$ (D) $16D$

MDT223

2. Find the value of $\begin{vmatrix}13 & 3 & 23\\ 30 & 7 & 53\\ 39 & 9 & 70\end{vmatrix}$

(A) 1 (B) 3 (C) 5 (D) 8

MDT224

3. The determinant $\begin{vmatrix}a & b & c\\ x & y & z\\ p & q & r\end{vmatrix}$ is same as:

(A) $\begin{vmatrix}a & b & p\\ y & b & q\\ z & c & r\end{vmatrix}$ (B) $\begin{vmatrix}x & b & q\\ y & a & p\\ z & c & r\end{vmatrix}$ (C) $\begin{vmatrix}x & z & y\\ p & r & q\\ a & c & b\end{vmatrix}$ (D) $\begin{vmatrix}y & b & q\\ x & a & p\\ z & c & r\end{vmatrix}$

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MDT225

4. If $A+B+C=\pi$, then the value of the determinant $D=\begin{vmatrix}\sin^2 A & \cot A & 1\\ \sin^2 B & \cot B & 1\\ \sin^2 C & \cot C & 1\end{vmatrix}$ is equal to

(A) 1 (B) –1 (C) 0 (D) none of these

MDT226

5. If $\begin{vmatrix}a+1 & a+2 & a+p\\ a+2 & a+3 & a+q\\ a+3 & a+4 & a+r\end{vmatrix}=0$, then p, q, r are in:

(A) A.P. (B) G.P. (C) H.P. (D) None of these

MDT227

Integer Type Question

6. Find the value of $\begin{vmatrix}53 & 106 & 159\\ 52 & 65 & 91\\ 102 & 153 & 221\end{vmatrix}$.

MDT228

For More Join: @IITJEE_Advanced

7. Solve for x : $\begin{vmatrix} x & 2 & 0 \\ 2+x & 5 & -1 \\ 5-x & 1 & 2 \end{vmatrix} = 0$

MDT229

8. If $D_r = \begin{vmatrix} 2r & 1 & n \\ 1 & -2 & 3 \\ 3 & 2 & 1 \end{vmatrix}$, then find the value of $\sum_{r=1}^{n} D_r$.

MDT230

9. If $D_r = \begin{vmatrix} 2^{r-1} & 2\left(3^{r-1}\right) & 4\left(5^{r-1}\right) \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$ then the value of $\sum_{r=1}^{n} D_r$ is

MDT231

Subjective Type Question

10. Without expanding the determinant prove that $\begin{vmatrix} a & p & \ell \\ b & q & m \\ c & r & n \end{vmatrix} + \begin{vmatrix} r & n & c \\ q & m & b \\ p & \ell & a \end{vmatrix} = 0$

MDT232

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	C	A	D	C	A	0	2	0	0.00	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-03

Single Correct Question

1. $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is equals to

(A) $(a-b)(b-c)(c-a)(ab+bc+ca)$ (B) $(a-b)(b-c)(c-a)$
(C) $(ab+bc+ca)$ (D) None of these

MDT233

2. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ is equals to

(A) $(a-b)(b-c)(c-a)(ab+bc+ca)$ (B) $(a-b)(b-c)(c-a)$
(C) $(a-b)(b-c)(c-a)(a+b+c)$ (D) None of these

MDT234

3. $\begin{vmatrix} a & b+c & a^2 \\ b & c+a & b^2 \\ c & a+b & c^2 \end{vmatrix}$ is equals to

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(A) $(a-b)(b-c)(c-a)$ (B) $-(a+b+c)(a-b)(b-c)(c-a)$
(C) $(a-b)(b-c)(c-a)(a+b+c)$ (D) None of these

MDT235

4. If $D_1 = \begin{vmatrix} ab^2-ac^2 & bc^2-a^2b & a^2c-b^2c \\ ac-ab & ab-bc & bc-ac \\ c-b & a-c & b-a \end{vmatrix}$ & $D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$, then D_1D_2 is equal to -

(A) 0 (B) D_1^2 (C) D_2^2 (D) D_2^3

MDT236

5. Factorize : $\begin{vmatrix} 3 & a+b+c & a^3+b^3+c^3 \\ a+b+c & a^2+b^2+c^2 & a^4+b^4+c^4 \\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^5+b^5+c^5 \end{vmatrix}$

(A) $[(a-b)^2(b-c)^2(c-a)^2(a+b+c)]$ (B) $[(a-b)(b-c)(c-a)(a+b+c)]$
(C) $[(a-b)^2(b-c)^2(c-a)^2]$ (D) None of these

MDT237

6. If $\begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix} = 3$, then find the value of $\begin{vmatrix} x^3-1 & 0 & x-x^4 \\ 0 & x-x^4 & x^3-1 \\ x-x^4 & x^3-1 & 0 \end{vmatrix}$

(A) 5 (B) 8 (C) 9 (D) 3

MDT238

For More Join: @IITJEE_Advanced

7. $\begin{vmatrix} a^2+b^2+c^2 & bc+ca+ab & bc+ca+ab \\ bc+ca+ab & a^2+b^2+c^2 & bc+ca+ab \\ bc+ca+ab & bc+ca+ab & a^2+b^2+c^2 \end{vmatrix}$ is always

(A) non-negative (B) 0 (C) negative (D) Can't say

MDT239

Subjective Type Question

8. Without expanding the determinant prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

MDT240

9. Using factor theorem, find the solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

MDT241

10. If the determinant $D = \begin{vmatrix} 1 & 1 & 1 \\ \alpha+\beta & \alpha^2+\beta^2 & 2\alpha\beta \\ \alpha+\beta & 2\alpha\beta & \alpha^2+\beta^2 \end{vmatrix}$ and $D_1 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{vmatrix}$, then find the determinant D_2 such that $D_2 = \frac{D}{D_1}$.

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MDT242

ANSWER KEY

Question	1	2	3	4	5	6	7	9	10
Answer	A	C	B	D	A	C	A	$(x = -1, 2)$	$\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \beta & \alpha \end{vmatrix}$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-04

Single Correct Question

1. $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$ is equal to

(A) $4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$ (B) $4(a-b)(b-c)(c-a)$

(C) $4\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ (D) all of these

MDT243

2. Which of the following determinant does not vanish?

(A) $\begin{vmatrix} 4 & 1 & 2 \\ 10 & 2 & 5 \\ 3 & 4 & 3/2 \end{vmatrix}$ (B) $\begin{vmatrix} 1 & 1 & 1 \\ \sin^2\theta & \sin^2 2\theta & \sin^2 3\theta \\ \cos^2\theta & \cos^2 2\theta & \cos^2 3\theta \end{vmatrix}$

(C) $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & -2 & -3 \end{vmatrix}$ (D) $\begin{vmatrix} \log 2 & \log 3 & -\log 6 \\ i & -2i & i^{2009} \\ 2+\sqrt{3} & 1+\sqrt{12} & -3-\sqrt{27} \end{vmatrix}$

MDT244

3. If $\Delta = \begin{vmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix}$, then for all $\theta \in R$

(A) $0 \le \Delta \le 2$ (B) $0 \le \Delta \le 1$ (C) $2 \le \Delta \le 4$ (D) $2 \le \Delta \le 6$

MDT245

4. The value of the determinant $\begin{vmatrix} \sin^2 13^\circ & \sin^2 77^\circ & \tan 135^\circ \\ \sin^2 77^\circ & \tan 135^\circ & \sin^2 13^\circ \\ \tan 135^\circ & \sin^2 13^\circ & \sin^2 77^\circ \end{vmatrix}$ is equal to

(A) – 1 (B) 0 (C) 3 (D) 7

MDT246

Rankers Academy JEE
For More Join: @IITJEE_Advanced

5. The determinant $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ is same as:

(A) $\begin{vmatrix} a & b & p \\ y & b & q \\ z & c & r \end{vmatrix}$ (B) $\begin{vmatrix} x & b & q \\ y & a & p \\ z & c & r \end{vmatrix}$ (C) $\begin{vmatrix} x & z & y \\ p & r & q \\ a & c & b \end{vmatrix}$ (D) $\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

MDT247

6. Let $A = \begin{bmatrix} \cos^{-1} x & \cos^{-1} y & \cos^{-1} z \\ \cos^{-1} y & \cos^{-1} z & \cos^{-1} x \\ \cos^{-1} z & \cos^{-1} x & \cos^{-1} y \end{bmatrix}$ such that $|A| = 0$, then maximum value of $x + y + z$ is

(A) 3 (B) 0 (C) 1 (D) 2

MDT248

7. If α, β & γ are the roots of the equation $x^3 + px + q = 0$, then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} =$

(A) p (B) q (C) $p^2 - 2q$ (D) none

MDT249

8. If a, b & c are non-zero real numbers then $D = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix} =$

Rankers Academy JEE

(A) abc (B) $a^2 b^2 c^2$ (C) $bc + ca + ab$ (D) zero

MDT250

9. The determinant $\begin{vmatrix} b_1 + c_1 & c_1 + a_1 & a_1 + b_1 \\ b_2 + c_2 & c_2 + a_2 & a_2 + b_2 \\ b_3 + c_3 & c_3 + a_3 & a_3 + b_3 \end{vmatrix} =$

(A) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (B) $2\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (C) $3\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (D) none of these

MDT251

10. The determinant $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix}$ is:

(A) 0
(B) independent of θ
(C) independent of ϕ
(D) independent of θ & ϕ both

MDT252

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	C	B	D	A	D	D	B	B

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-05

Single Correct Question

1. The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is

(A) $k(a + b)(b + c)(c + a)$
(B) $kabc(a^2 + b^2 + c^2)$
(C) $k(a - b)(b - c)(c - a)$
(D) $k(a + b - c)(b + c - a)(c + a - b)$

MDT253

2. Let a, b, c be such that $b(a + c) \neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$, then the value of n is :-

(A) Any odd integer
(B) Any integer
(C) Zero
(D) Any even integer

MDT254

Rankers Academy JEE

3. $\Delta = \begin{vmatrix} 1+a^2+a^4 & 1+ab+a^2b^2 & 1+ac+a^2c^2 \\ 1+ab+a^2b^2 & 1+b^2+b^4 & 1+bc+b^2c^2 \\ 1+ac+a^2c^2 & 1+bc+b^2c^2 & 1+c^2+c^4 \end{vmatrix}$ is equal to

(A) $(a - b)^2 (b - c)^2 (c - a)^2$
(B) $2(a - b) (b - c) (c - a)$
(C) $4(a - b) (b - c) (c - a)$
(D) $(a + b + c)^3$

MDT255

4. Value of the $\Delta = \begin{vmatrix} a^3-x & a^4-x & a^5-x \\ a^5-x & a^6-x & a^7-x \\ a^7-x & a^8-x & a^9-x \end{vmatrix}$ is

(A) 0
(B) $(a^3 - 1) (a^6 - 1) (a^9 - 1)$
(C) $(a^3 + 1) (a^6 + 1) (a^9 + 1)$
(D) $a^{15} - 1$

MDT256

Integer Type Question

5. Find the value of the determinant $\begin{vmatrix} a^2+b^2 & a^2-c^2 & a^2-c^2 \\ -a^2 & 0 & c^2-a^2 \\ b^2 & -c^2 & b^2 \end{vmatrix}$.

MDT257

For More Join: @IITJEE_Advanced

6. If $\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} \geq 0$, where $a, b, c \in R^+$, then $\frac{a+b}{c}$ is

MDT258

7. If $\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = k(\alpha a + \beta b + \gamma c)^3$, then $(2\alpha + \beta - \gamma)^k$ is $(\alpha, \beta, \gamma, k \in z^+)$

MDT259

8. If $\begin{vmatrix} -bc & b^2+bc & c^2+bc \\ a^2+ac & -ac & c^2+ac \\ a^2+ab & b^2+ab & -ab \end{vmatrix} = 64$, then $(ab + bc + ac)$ is :

MDT260

9. If $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N+1 \end{vmatrix}$ and $\sum_{n=1}^{N} U_n = \lambda \sum_{n=1}^{N} n^2$, then λ is

MDT261

Subjective Type Question

10. Prove that $\begin{vmatrix} a & b & c \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$.

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MDT262

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	C	A	A	A	0	2	4	4	2	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Determinants : RACE-06

Single Correct Question

1. The product of all the values of α for which equations $(a-\alpha)x + by + c = 0$, $cx + (a-\alpha)y + b = 0$ and $bx + cy + a - \alpha = 0$ are consistent, when $a, b, c > 0$, is -

(A) zero (B) positive (C) negative (D) one

MDT263

2. The number of values of k for which the system of equations

$(k+1)x + 8y = 4k$

$kx + (k+3)y = 3k - 1$

has infinitely many solutions is

(A) 0 (B) 1 (C) 2 (D) Infinite

MDT264

3. If the system of equations $x + y + z = 2$, $2x + y - z = 3$ & $3x + 2y + kz = 4$ has a unique solution, then

(A) $k \neq 0$ (B) $-1 < k < 1$ (C) $-2 < k < 1$ (D) $k = 0$

MDT265

Rankers Academy JEE

4. The system of equations $\lambda x + y + z = 0$, $-x + \lambda y + z = 0$ & $-x - y + \lambda z = 0$ has a non-trivial solution, then possible values of λ are -

(A) 0 (B) 1 (C) -3 (D) $\sqrt{3}$

MDT266

5. Consider the system of linear equations :

$x_1 + 2x_2 + x_3 = 3, 2x_1 + 3x_2 + x_3 = 3, 3x_1 + 5x_2 + 2x_3 = 1$

The system has

(A) Infinite number of solutions (B) Exactly 3 solutions

(C) A unique solution (D) No solution

MDT267

6. The number of values of k for which the linear equations $4x + ky + 2z = 0$, $kx + 4y + z = 0$, $2x + 2y + z = 0$ possess a non-zero solution is :

(A) 1 (B) zero (C) 3 (D) 2

MDT268

7. If the trivial solution is the only solution of the system of equations

$x - ky + z = 0$, $kx + 3y - kz = 0$, $3x + y - z = 0$ Then the set of all values of k is :

(A) $\{2, -3\}$ (B) $R - \{2, -3\}$ (C) $R - \{2\}$ (D) $R - \{-3\}$

MDT269

8. Solve the following system of equations $x + y + z = 5$, $2x + 2y + 2z = 7$, $3x + 3y + 3z = 6$

(A) Unique solution (B) infinite solution (C) No solution (D) None of these

MDT270

For More Join: @IITJEE_Advanced

9. The system of equation

$-kx + 3y - 14z = 25$

$-15x + 4y - kz = 3$

$-4x + y + 3z = 4$

is consistent for all k in the set

(A) R (B) $R - \{-11,13\}$ (C) $R - \{13\}$ (D) $R - \{-11,11\}$

MDT271

Multiple Correct Question

10. The values of 'λ' for which the system of equations

$(1 - \lambda)x + 3y - 4z = 0$

$x - (3 + \lambda)y + 5z = 0$

$3x + y - \lambda z = 0$

possesses non-trivial solution, is/are

(A) –1 (B) 0 (C) 1 (D) 2

MDT272

Subjective Type Question

11. Find nature of solution for given system of equations

$2x + y + z = 3$; $x + 2y + z = 4$; $3x + z = 2$

MDT273

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	A	A	D	D	B	C	D	A,B

11. (Infinite Solutions)

For More Join: @IITJEE_Advanced